

MODULE 1

3 MARKS

1. List any two purposes of recommender systems.
2. With an example, explain working of Recommendation Algorithm?
3. Indicate the main idea of collaborative recommendation approaches.
4. Discuss the cases in which content-based recommendations will not perform as well as collaborative filtering.
5. How the data can be pre-processed for item based filtering?
6. Write a note on collaborative recommendation?
7. How is a collaborative recommendation system categorized?
8. What inquiries will be tackled regarding the recent development in recommendation systems?
9. What is the central concept behind collaborative recommendation?
10. Describe the approaches for implicit and explicit ratings in collaboration system.
- 11.

7MARKS

1. Explain about Rocchio's relevance feedback method
2. Summarize the implicit and explicit rating mechanism in collaborative recommendation approaches
3. Describe about user-based nearest neighbour recommendation system which deals with new items for which no ratings exist.
4. Elaborate on the user-based nearest neighbour recommendation system with an example.

14 MARKS

1. Describe the two types of outputs generated with pure collaborative approaches that takes matrix of given user–item ratings as the only input?
2. Discuss content-based recommendation, in light of explicitly asking the user for his or her interests?
3. Write a note on cosine- similarity and what is the application of cosine- similarity in Recommendation system?
4. Explain similarity based retrieval and other text classification methods?

MODULE 2

3 MARKS

1. Define classical constraint satisfaction problem.
2. In what sense is Critiquing meant?
3. Differentiate between Dependent default and derived default?
4. How can the performance of a vector space model be enhanced through various techniques and optimizations?
5. Explain TF-IDF.

6. Explain knowledge based recommendation.
- 7.

7 MARKS

1. How can we deal unsatisfiable requirements and empty results sets?
2. Explain QuickXPlain algorithm that calculates one conflict set at a time for a given set of constraints.
3. Explain about the ranking of items/utility-based recommendation.
4. How can we deal un-satisfiable requirements and empty results sets?
5. How defaults are utilized in constraint based recommendation?

14 MARKS

1. Describe Critiquing algorithms.
2. How we can summarize the general interaction flow of knowledge based, conversational recommendation system?

MODULE 3

3 marks

1. What is meant by monolithic hybridization design?
2. Differentiate between parallelized hybridization design and pipelined hybridization design with suitable diagram.
3. What are the limitations of hybridization strategies?
4. Explain about feature combination hybrids
5. Can we treat recommendation problem as utility function?
6. What are the two dimensions of hybrid recommendation system?

7 marks

1. Explain about feature combination hybrids.
2. Describe feature augmentation hybrid.
3. Explain about different parallelized hybridization strategies.
4. Describe pipelined hybridization design methods.
5. Explain recommendation paradigm?
6. Describe feature augmentation hybrids.
7. Explain about different parallelized hybridization strategies.
8. Differentiate between cascade hybrids and meta- level hybrids?

MODULE 4

3 MARKS

1. What is Impact of long tail?
2. Explain evaluation paradigms.
3. Differentiate RMSE versus MAE?
4. How can we distinguish between confidence and trust?
5. Explain A/B testing.

7 MARKS

1. Discuss about the design issues in offline recommender evaluation. Illustrate with a case study.
2. Explain about accuracy metrics in offline evaluation
3. Explain about offline and online evaluations in recommender systems
4. Discuss about the various evaluation paradigms in recommendation.

14 MARKS

1. Discuss about the general goals of evaluation design?
2. Discuss about the design issues in offline recommender evaluation. Illustrate with a case study.
3. Explain the Accuracy metrics in offline evaluation.

MODULE 5

3MARKS

1. What is product push attack and nuke attack?
2. Specify the effect of base recommendation algorithm.
3. What is product push attack and nuke attack?
4. Illustrate different methods used to detect attacks on existing recommendation system.
5. Discuss about different attacks on recommender system?
6. What are the different methods in group attack profile detection?

7 MARKS

1. Discuss about different methods available to detect attacks on recommender system
2. Explain how to design robust recommendation algorithms.

14 MARKS

1. Discuss about different attacks on recommender system.
2. How do you quantify attack impact on recommender system?